Rustem Islamov

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Education

PhD in Computer Science — Oct. 2023 – Present
University of Basel, Supervisor: Aurelien Lucchi — Basel, Switzerland
PhD Research: Optimization for Machine Learning

Master of Science in Applied Mathematics — Sept. 2021 – Aug. 2023
Institut Polytechnique de Paris — Palaiseau, France
GPA: 17.65/20, transcript of records
Supervisor: Olivier Fercoq, Thesis supervisor: Dan Alistarh
Thesis: Distributed Second Order Methods with Fast Rates and Compressed Communication

Bachelor of Science in Applied Mathematics and Physics — Sept. 2017 – July 2021
Moscow Institute of Physics and Technology — Dolgoprudny, Russia
Supervisor: Peter Richtárik
GPA: 4.95/5 (9.27/10), Top 3 at the department, transcript of records
Thesis: Unified Analysis of Asynchronous Algorithms

Research Interests and Technical Skills

Optimization (deep learning optimizers, scaling laws, differential privacy, computation- and memory-efficient methods), Deep Learning (loss landscape of neural networks)

Programming Languages: Python (NumPy, Matplotlib, PyTorch, Pandas), C++, LaTeX
Mathematics: Calculus, Linear Algebra, Probability Theory, Convex Analysis, Deep Learning

Internships and Research Visits

Research Visit in the group of Prof. Eduard Gorbunov — Dec. 2025 – Feb. 2026
MBZUAI, Host: Eduard Gorbunov — Abu Dhabi, UAE

Internship at Distributed Algorithms and Systems Lab — Apr. 2023 – Oct. 2023
IST Austria, Supervisors: Mher Safaryan, Dan Alistarh — Klosterneuburg, Austria

Internship at Machine Learning and Optimization Lab — Apr. 2022 – Aug. 2022
EPFL, Supervisors: Hadrien Hendrikx, Martin Jaggi — Lausanne, Switzerland

Internship at Optimization and Machine Learning Lab — Mar. 2021 – Aug. 2021
KAUST, Supervisor: Peter Richtárik — Jul. 2020 – Dec. 2020
Thuwal, Saudi Arabia

Scholarships, Honors and Awards

Top Reviewer for ICML 2025-2026, NeurIPS 2025-2026, ICLR 2026 — 2025-2026
French Embassy Scholarship — Sept. 2022 – May. 2023
PhD Track Excellence Scholarship — Sept. 2021 – Mar. 2023
Increased Abramov Excellence Scholarship — Sept. 2017 – June 2020

All Publications (19 Conference and 3 Journal Papers)

25. **AdamW Scaling Laws from First Principles** — Preprint
R. Islamov*, I. Mavrothalassitis*, V. Boreiko*, A. Silveti-Falls, E. Gorbunov, V. Cevher, 2026.

24. **What's in a Smoothness Constant? Tighter Rates for Local SGD with Bounded Second-order Heterogeneity** — NeurIPS 2026 — arXiv
K. Patel*, R. Islamov*, S. Stich, A. Lucchi, E. Gorbunov, L. Wang, 2026

23. **Beyond a Single Explanation of the Adam-SGD Gap** — NeurIPS 2026 — arXiv
C. Zhang*, R. Islamov*, E.M. Compagnoni, J. Pang, A. Lucchi, A. Orvieto, 2026

22. **Byzantine-Robust and Differentially Private Federated Optimization under Weaker Assumptions** — UAI 2026
R. Islamov, G. Malinovsky, A. Gaponov, A. Lucchi, P. Richtárik, E. Gorbunov, 2026.

21. **On the Role of Batch Size in Stochastic Conditional Gradient Methods** ICML 2026
R. Islamov, R. Machacek, A. Lucchi, A. Silveti-Falls, E. Gorbunov, V. Cevher, 2026.

20. **Non-Euclidean Gradient Descent Operates at the Edge of Stability** Oral ICML 2026
R. Islamov, M. Crawshaw, J. Cohen, R. Gower, 2026.

19. **Adaptive Methods Are Preferable in High Privacy Settings: An SDE Perspective** ICLR 2026
E. M. Compagnoni, A. Stanghellini, R. Islamov, A. Lucchi, A. Koloskova, 2026.

18. **Why Do We Need Warm-up? A Theoretical Perspective** ICML 2026
F. Alimisis*, R. Islamov*, A. Lucchi, 2025.

17. **Enhancing Optimizer Stability: Momentum Adaptation of NGN Step-size** NeurIPS 2025
R. Islamov, N. Ajroldi, A. Orvieto, A. Lucchi, 2025

16. **On the Interaction of Noise, Compression Role, and Adaptivity under (L_0, L_1)-Smoothness: An SDE-based Approach** ICML 2026
E. M. Compagnoni, R. Islamov, A. Orvieto, E. Gorbunov, 2025.

15. **Safe-EF: Error Feedback for Non-smooth Constrained Optimization** ICML 2025
R. Islamov, Y. As, I. Fatkhullin, 2025.

14. **Double Momentum and Error Feedback for Clipping with Fast Rates and Differential Privacy** Preprint arXiv
R. Islamov, S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov, 2025

13. **Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs** Oral AISTATS 2025
E. M. Compagnoni, R. Islamov, F. N. Proske, A. Lucchi, 2025.

12. **Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise** ICLR 2025
E. M. Compagnoni, T. Liu, R. Islamov, F. N. Proske, A. Orvieto, A. Lucchi, 2024.

11. **Loss Landscape Characterization of Neural Networks without Over-Parametrization** NeurIPS 2024
R. Islamov, N. Ajroldi, A. Orvieto, A. Lucchi, 2024.

10. **Towards Faster Decentralized Stochastic Optimization with Communication Compression** ICLR 2025
R. Islamov*, Y. Gao*, S. Stich, 2024.

9. **EControl: Fast Distributed Optimization with Compression and Error Control** ICLR 2024
Y. Gao*, R. Islamov*, S. Stich, 2023.

8. **AsGrad: A Sharp Unified Analysis of Asynchronous-SGD Algorithms** AISTATS 2024
R. Islamov, M. Safaryan, D. Alistarh, 2023.

7. **Partially Personalized Federated Learning: Breaking the Curse of Data Heterogeneity** TMLR 2023
K. Mishchenko, R. Islamov, E. Gorbunov, S. Horváth, 2023.

6. **Clip21: Error Feedback for Gradient Clipping** Preprint arXiv
S. Khirirat, E. Gorbunov, S. Horváth, R. Islamov, F. Karray, P. Richtárik, 2023.

5. **Adaptive Compression for Communication-Efficient Distributed Training** TMLR 2022
M. Makarenko, E. Gasanov, R. Islamov, A. Sadiev, P. Richtárik, 2022.

4. **Distributed Newton-Type Methods with Communication Compression and Bernoulli Aggregation** TMLR 2022
R. Islamov, X. Qian, S. Hanzely, M. Safaryan, P. Richtárik, 2022.

3. **Basis Matters: Better Communication-Efficient Second Order Methods for Federated Learning** AISTATS 2022
   X. Qian, R. Islamov, M. Safaryan, P. Richtárik, 2022.
2. **FedNL: Making Newton-Type Methods Applicable to Federated Learning** ICML 2022
   M. Safaryan, R. Islamov, X. Qian, P. Richtárik, 2022.
1. **Distributed Second Order Methods with Fast Rates and Compressed Communication** ICML 2021
   R. Islamov, X. Qian, P. Richtárik, 2021.

Reviewing and Teaching

Teaching Assistant for Continuous Optimization course — Spring Semesters 2024-2026
Reviewer for Transactions on Machine Learning Research TMLR — 2023-2026
Reviewer for Conference on Neural Information Processing Systems NeurIPS — 2024-2025
Reviewer for International Conference on Machine Learning ICML — 2024-2026
Reviewer for International Conference on Learning Representations ICLR — 2024-2027
Reviewer for Artificial Intelligence and Statistics Conference AISTATS — 2023-2026
Reviewer for Journal on Machine Learning Research JMLR — 2024-2026
Reviewer for Journal of Parallel and Distributed Computing JPDC — 2024
Reviewer for Journal of Optimization Theory and Applications JOTA — 2023

Talks and Presentations

Talk at Swiss Optimization Symposium, Links: slides — 23 August, 2026
Lightning Talk at Protocol Learning: Decentralized Collaborative Learning at Scale (ICML 2026 Workshop) Links: paper — 10 July, 2026
Oral Presentation at ICML 2026 Conference, Links: paper — 9 July, 2026
Invited Talk at SIAM OP26 Conference, Links: paper — 2-5 June, 2026
Invited Talk at EUROPT Conference, Links: paper — 29 June-2 July, 2025
Invited Talk at Rising Stars in AI Symposium at KAUST workshop, Links: paper, slides — 21 February, 2024
Talk at NTDS workshop, Links: paper — 24 October, 2023
Invited Talk at CISPA, Links: paper — 16 March, 2023
Talk at ETH AI Center Symposium for PhD fellows, Links: paper — 9-10 February, 2023
Talk at KAUST Conference on Artificial Intelligence, Links: video, paper — 28 April, 2021

Languages

Russian: Native
English: Advanced (C1)
French: Elementary (A2)

Hobbies and Interests

Football, former member of student football team
Traveling, hiking, photo shooting